

Flow Assurance — Practical Study Guide

A field-oriented reference covering the core engineering topics in flow assurance for subsea and long-distance pipeline systems — combining industry-standard theory with worked sample calculations, sample datasheets, and design-basis assumptions drawn from real project execution.

Illustrative project used throughout this guide: a subsea gas-condensate tieback (wellhead to host platform), with produced water, sand potential, and a compositionally rich fluid prone to hydrate, wax, and CO₂ corrosion risk. All numbers below are worked sample calculations for study purposes — always replace with project-specific PVT, water chemistry, and flow simulation data.

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1. Design Basis & Assumptions

Flow assurance basis is normally issued as a "**Flow Assurance Design Basis & Philosophy**" document, frozen before pipeline thermal design, chemical injection sizing, and dynamic (OLGA/PIPESIM) simulation begin. Revisiting fluid composition, water cut, or thermal targets mid-project is a major source of rework — it changes insulation thickness, chemical injection tie-in sizing, and sometimes pipeline metallurgy.

1.1 Field & Fluid Basis (illustrative project)

Parameter	Value	Notes
Tieback length	25 km	Wellhead to host platform
Water depth	1,200 m	Seabed ambient $\approx 4\text{ }^{\circ}\text{C}$ ($39\text{ }^{\circ}\text{F}$) year-round
Reservoir fluid type	Gas-condensate, rich in C2–C5	GOR $\approx 15,000\text{ scf/bbl}$
Flowing wellhead pressure/temperature	3,500 psia / $160\text{ }^{\circ}\text{F}$ ($71\text{ }^{\circ}\text{C}$)	Declines with reservoir depletion — check end-of-field-life case too
Arrival pressure/temperature at host (normal flow)	1,500 psia / $140\text{ }^{\circ}\text{F}$ ($60\text{ }^{\circ}\text{C}$)	—
Produced water cut (initial $\rightarrow$ late life)	5% $\rightarrow$ 40%	Late-life case often governs hydrate/scale/corrosion risk
CO ₂ content	5 mol%	Governs corrosion basis
H ₂ S content	50 ppm	Sour service screening required (NACE MR0175/ISO 15156)
Sand production potential	Trace, screened wells	Erosion basis uses a low but non-zero sand rate
Wax Appearance Temperature (WAT)	$100\text{ }^{\circ}\text{F}$ ($37.8\text{ }^{\circ}\text{C}$)	From lab (cold-finger/DSC) testing on live/dead oil
Asphaltene onset pressure (AOP)	$\sim 2,800\text{ psia}$ at reservoir temp	From lab PVT/gravimetric titration

1.2 Codes & Standards / Methodology Basis

- **Hydrate:** Multiflash/PVTsim or CSMHyd-type thermodynamic hydrate modeling for the equilibrium curve; **Hammerschmidt equation** for quick-check inhibitor dosage (Section 8.2)
- **Wax:** Lab WAT/pour point testing; steady-state and transient thermal simulation (OLGA/PIPESIM) for cooldown; **lumped-capacitance screening calc** for first-pass insulation sizing (Section 8.3)
- **Asphaltene:** Lab asphaltene onset pressure (AOP) envelope; compositional simulation for pressure-depletion trajectory

- **Scale:** Produced/injection water compatibility studies; saturation ratio / scaling indices (Stiff-Davis, Langelier, or direct ion-product-vs-Ksp screening, Section 8.6)
- **Corrosion:** NORSOK M-506 or de Waard-Milliams CO₂ corrosion prediction models; NACE MR0175/ISO 15156 for sour service material selection
- **Erosion:** API RP 14E erosional velocity guideline (screening); DNV RP O501 for detailed sand-erosion rate prediction
- **Multiphase/transient:** OLGA, PIPESIM, LedaFlow, or CFD for slugging, restart, and blowdown transient prediction

1.3 Key Design Assumptions (typical, to be confirmed per project)

Assumption	Typical Value	Basis / Justification
Hydrate subcooling safety margin	3–5 °F (1.5–3 °C) additional beyond the calculated suppression requirement	Company philosophy; accounts for model/measurement uncertainty
Methanol density	0.792 kg/L	Standard reference value
Required no-flow (shut-in) protection time before wax/hydrate risk	6 hours (planned), longer for unplanned per project philosophy	Confirm with Operations — drives insulation/active heating basis
API RP 14E erosional velocity constant, C	100 (continuous service, solids present)	API RP 14E; C=125–150 sometimes used for solids-free/intermittent service — confirm basis
CO ₂ corrosion inhibitor efficiency (design)	90–95%	Vendor qualification testing required; do not assume without data
Corrosion allowance	3 mm	Project material spec
Design life	25 years	Project spec
Scale inhibitor squeeze lifetime (design)	6–12 months between treatments	Confirm with produced water forecast and well intervention plan

⚠ Practical note: Every flow assurance discipline above interacts with the others — insulation added for wax control also slows the hydrate cooldown clock (helps), but a corrosion-inhibitor film can be stripped by high-velocity erosive flow (hurts). Always review the design basis as an integrated package, not as six independent checklists.

2. Hydrate Formation

2.1 Cause

Water + light hydrocarbon gas at **high pressure and low temperature** forms ice-like crystalline structures (clathrates) that can plug pipelines, chokes, and subsea equipment — even at temperatures well above 0 °C, since hydrate formation is pressure-dependent, not just temperature-dependent.

2.2 Prevention Methods

Method	Mechanism	Typical Use
Thermodynamic inhibitors (methanol, MEG)	Shift the hydrate equilibrium curve to lower temperature/higher pressure (freezing-point-depression-like effect)	Continuous injection or batch dosing for shut-in/restart protection
Low-dosage hydrate inhibitors (LDHI)	Kinetic inhibitors (delay nucleation) or anti-agglomerants (prevent crystal bonding) — effective at much lower dose than thermodynamic inhibitors	Systems with moderate subcooling where full thermodynamic inhibition is uneconomical
Insulation and heating	Keep fluid temperature above the hydrate curve for as long as practical (passive insulation) or indefinitely (active heating/electrical trace heating, hot-water circulation)	Standard for subsea tiebacks; often combined with chemical injection for shut-in events beyond insulation's no-flow protection time

2.3 Key Calculations

- **Hammerschmidt equation** — quick-check inhibitor dosage for a required hydrate suppression temperature (Section 8.2).
- **Subcooling margin analysis** — the difference between the hydrate equilibrium temperature (at operating pressure) and the actual fluid temperature; a positive subcooling margin (fluid colder than the hydrate curve) indicates the system is inside the hydrate-risk region (Section 8.1).

3. Wax Deposition

3.1 Cause

Crude oil cools below its **Wax Appearance Temperature (WAT)** — the temperature at which dissolved paraffins begin crystallizing out of solution — leading to wax deposition on cold pipe walls, progressively restricting flow area.

3.2 Controls

Method	Notes
Pour-point depressants (PPD)	Modify wax crystal structure to reduce pour point / improve low-temperature flowability; do not eliminate deposition risk on their own
Wax inhibitors	Reduce deposition rate/adhesion; efficacy is fluid-specific — always lab-test with the actual (or representative) crude
Pigging	Mechanical removal of accumulated wax; pigging frequency is a key operating parameter derived from deposition-rate modeling
Insulation	Keeps fluid temperature above WAT for longer, both during normal flow (steady-state heat loss) and during shut-in (transient cooldown) — see Calc Sheet 8.3

3.3 Design Focus

Thermal management is the primary design lever for subsea tiebacks and long pipelines — insulation thickness/type (wet insulation, pipe-in-pipe, or active heating) is sized against both the **steady-state** arrival temperature target and the **transient no-flow cooldown time** requirement (Calc Sheet 8.3 works through both).

4. Asphaltene Precipitation

4.1 Cause

Asphaltenes are colloiddally stabilized heavy fractions in crude oil; they precipitate when that stability is disrupted by:

- **Pressure drop** — most commonly, crossing below the **Asphaltene Onset Pressure (AOP)** during production (pressure depletion can move the operating point through the unstable region before eventually re-stabilizing at very low pressure)
- **Blending** — mixing crudes of different asphaltene stability (common risk when commingling production streams or during tanker blending)
- **Composition changes** — gas injection (miscible EOR), CO₂ breakthrough, or changing GOR over field life

4.2 Controls

Method	Notes
Asphaltene inhibitors	Chemical dispersants/stabilizers, typically continuously injected downhole or at the wellhead; must be lab-qualified against the specific crude — poor inhibitor selection can <i>promote</i> rather than prevent precipitation
Dispersants	Keep precipitated particles suspended/small rather than allowing agglomeration and deposition
Pressure management	Choke/wellhead pressure management to avoid dwelling in the unstable pressure window where practical

4.3 Difference from Wax

	Wax	Asphaltenes
Primary driver	Temperature (crosses WAT)	Pressure/chemistry (crosses AOP, or composition/blending shift)
Typical deposit location	Cold spots — seabed sections, risers near mudline	Near-wellbore, tubing, chokes — often where the <i>largest pressure drop</i> occurs, not necessarily the coldest point
Reversibility	Can partially re-dissolve on reheating	Precipitation is not simply temperature-reversible; requires pressure/composition to return toward the stable region

5. Scale Formation

5.1 Cause

- **Produced water supersaturation** — as pressure/temperature change along the production path, dissolved mineral salts (commonly CaCO_3 , CaSO_4 , BaSO_4 , SrSO_4) can exceed their solubility limit and precipitate.
- **Mixing of incompatible brines** — most notably seawater (rich in sulfate) mixing with formation water (rich in barium/strontium) during seawater injection for pressure support, producing highly insoluble BaSO_4 / SrSO_4 scale.

5.2 Controls

- **Scale inhibitors** — continuous downhole/topside injection, or **squeeze treatments** (inhibitor is bullheaded into the near-wellbore formation and slowly produced back

over weeks-to-months, providing extended protection without continuous topside injection infrastructure).

- Compatibility testing (mixing produced and injection water at reservoir conditions) is essential before committing to a waterflood scheme — see Calc Sheet 8.6 for a simplified saturation-ratio screening approach.

5.3 Risk

Scale plugs valves, chokes, and subsea equipment — often at the point of maximum pressure drop (chokes, safety valves) where flashing/temperature change most rapidly triggers precipitation, similar in that respect to asphaltenes.

6. Corrosion & Erosion

6.1 Corrosion

- **Sweet corrosion (CO_2):** Carbonic acid formed from dissolved CO_2 attacks carbon steel; rate is a strong function of CO_2 partial pressure, temperature, pH, and flow regime (Calc Sheet 8.5 works a simplified de Waard-Milliams example).
- **Sour corrosion (H_2S):** Forms iron sulfide scale (can be locally protective or locally aggressive depending on conditions) and introduces **sulfide stress cracking (SSC)** risk for susceptible materials — governed by NACE MR0175/ISO 15156 material selection rules, not just a corrosion-rate calculation.
- **Controls:** Material selection (carbon steel + inhibitor vs. corrosion-resistant alloy (CRA) cladding/lining), continuous corrosion inhibitor injection, and corrosion monitoring (coupons, ER/LPR probes, intelligent pigging).

6.2 Erosion

- High-velocity fluid carrying **sand or other solids** causes wall thinning, particularly at flow-direction changes (elbows, tees, chokes) where particle impingement is highest.
- **Controls:** Velocity limits (API RP 14E erosional velocity screening, Calc Sheet 8.4), pipe sizing (larger ID reduces velocity for the same volumetric flow), erosion-resistant materials/coatings, and blind-tee or long-radius-bend geometry at high-risk locations to reduce direct impingement.

⚠ **Practical note:** Corrosion inhibitor film integrity and erosion are linked — flow velocities high enough to be erosion-risk are often also high enough to mechanically strip a corrosion inhibitor film, so an erosion check that "passes" on its own is not sufficient evidence that corrosion is also under control at the same location.

7. Multiphase Flow & Transient Operations

7.1 Slugging

Severe pressure/flow fluctuations caused by intermittent liquid accumulation and sweep-out along a pipeline — commonly **terrain-induced** (low points in an undulating seabed profile), **riser-induced** (severe slugging at the base of a riser), or **hydrodynamic** (Kelvin-Helmholtz instability at gas-liquid interfaces in near-horizontal flow). Slug catchers at the receiving facility are sized against the predicted slug volume/frequency from dynamic multiphase simulation.

7.2 Restart/Shut-in Risks

- **Hydrate plugs during cold restart** are one of the highest-consequence flow assurance risks — a shut-in pipeline cools toward seabed ambient temperature over time (Calc Sheet 8.3 methodology), and if the no-flow duration exceeds the calculated/inhibited protection time, restart must proceed with a hydrate-management plan (e.g., depressurization before restart, or confirmed inhibitor coverage) rather than simply reopening the well.
- **Practical experience:** Depressurizing a shut-in subsea pipeline (rather than restarting under full pressure) is a common hydrate-avoidance strategy — but depressurization itself causes further Joule-Thomson cooling, so the depressurization rate/target must be checked against the hydrate curve too, not assumed to be automatically safe.

7.3 Modeling Tools

Tool	Typical Use
OLGA	Industry-standard transient multiphase flow simulator — slugging, cooldown, restart, pigging
PIPESIM	Steady-state and transient multiphase network simulation, well-to-facility
CFD	Detailed local flow prediction (e.g., erosion at a specific fitting, slug impact loading) where 1D network tools are insufficient

8. Sample Calculation Sheets

All calculations below use the illustrative project basis in Section 1. Values are for study purposes — verify against project-specific PVT/water-chemistry data and dynamic simulation.

8.1 Calc Sheet 1 — Hydrate Subcooling Margin

Given:

- Operating pressure at seabed cold spot, $P = 1,500$ psia

- Hydrate equilibrium temperature at this pressure (from thermodynamic hydrate model), $T_{\text{hyd}} = 65\text{ }^{\circ}\text{F}$
- Actual flowing fluid temperature at this location, $T_{\text{actual}} = 40\text{ }^{\circ}\text{F}$
- Required additional safety margin = $5\text{ }^{\circ}\text{F}$ (Section 1.3 basis)

Step 1 — Calculate subcooling (how far into the hydrate region the system currently sits):

$$\Delta T_{\text{subcool}} = T_{\text{hyd}} - T_{\text{actual}} = 65 - 40 = 25\text{ }^{\circ}\text{F}$$

A positive value means the fluid is **$25\text{ }^{\circ}\text{F}$ colder** than the hydrate formation temperature at this pressure — the system is well inside the hydrate-risk region without inhibition.

Step 2 — Determine target hydrate-curve temperature (with safety margin):

$$T_{\text{hyd,target}} = T_{\text{actual}} - \text{safety margin} = 40 - 5 = 35\text{ }^{\circ}\text{F}$$

Step 3 — Required hydrate suppression (how far the inhibitor must shift the curve):

$$\Delta T_{\text{required}} = T_{\text{hyd}} - T_{\text{hyd,target}} = 65 - 35 = 30\text{ }^{\circ}\text{F}$$

Result: The inhibitor system must be capable of suppressing the hydrate formation temperature by **$30\text{ }^{\circ}\text{F}$** at this location. This value feeds directly into the Hammerschmidt dosage calculation (Calc Sheet 8.2).

✦ **Assumption check:** T_{hyd} should come from a compositional thermodynamic hydrate model (e.g., Multiflash), not a generic chart — hydrate curves are sensitive to gas composition ($\text{CO}_2/\text{H}_2\text{S}$ content shifts the curve), and using a generic "sales gas" hydrate curve for a CO_2 -rich stream can significantly under- or over-state the real risk.

8.2 Calc Sheet 2 — Methanol Dosage (Hammerschmidt Equation)

Given (from Calc Sheet 8.1):

- Required hydrate suppression, $\Delta T = 30\text{ }^{\circ}\text{F}$
- Hammerschmidt constant for methanol, $K = 2,335$ ($^{\circ}\text{F}$ basis)
- Methanol molecular weight, $M = 32.04\text{ g/mol}$
- Produced water rate = 50 bbl/day (late-life case per Section 1.3 is higher — this is an early-life illustrative case; repeat for the late-life 40% water-cut case in a real design)
- Methanol density = 0.792 kg/L

Step 1 — Hammerschmidt equation, solved for required methanol wt% (W) in the free water phase:

$$\Delta T = (K \times W) / [M \times (100 - W)]$$

$$30 = (2,335 \times W) / [32.04 \times (100 - W)]$$

$$30 \times 32.04 \times (100 - W) = 2,335 \times W$$

$$961.2 \times (100 - W) = 2,335 W$$

$$96,120 - 961.2 W = 2,335 W$$

$$96,120 = 3,296.2 W$$

$$W \approx 29.16 \text{ wt\%}$$

Result: The water phase must contain **≈29.2 wt% methanol** to achieve the required 30 °F suppression.

Step 2 — Convert to mass injection rate:

$$\text{Water rate} = 50 \text{ bbl/day} \times 0.159 \text{ m}^3/\text{bbl} \times 1,000 \text{ kg/m}^3 \approx 7,950 \text{ kg/day}$$

$$W = M_{\text{methanol}} / (M_{\text{methanol}} + M_{\text{water}}) \times 100$$

$$0.2916 = M_{\text{methanol}} / (M_{\text{methanol}} + 7,950)$$

$$M_{\text{methanol}} = 0.2916 \times (M_{\text{methanol}} + 7,950)$$

$$M_{\text{methanol}} = 2,318.5 + 0.2916 M_{\text{methanol}}$$

$$0.7084 M_{\text{methanol}} = 2,318.5$$

$$M_{\text{methanol}} \approx 3,273 \text{ kg/day}$$

Step 3 — Convert to volumetric injection rate:

$$\text{Volume} = 3,273 \text{ kg/day} \div 0.792 \text{ kg/L} \approx 4,133 \text{ L/day} \approx 26 \text{ bbl/day}$$

Result: Required methanol injection rate **≈ 3.27 tonnes/day (≈26 bbl/day)** for this early-life water cut. This sizes the methanol storage, injection pump, and topside/umbilical tie-in capacity.

✦ **Assumption check:** This must be re-run for the **late-life water cut (40%)** per the design basis — at higher water cut, the same wt% target requires proportionally more methanol mass, and this late-life case is very often the one that actually governs storage/pump/umbilical sizing, not the early-life case shown here. Also confirm whether MEG (regenerable, lower OPEX for high water-cut systems) is more economical than once-through methanol for the late-life case — this is a common flow assurance economic trade-off study.

8.3 Calc Sheet 3 — Wax Cooldown / Insulation Screening (Lumped Capacitance Method)

Given:

- Fluid arrival temperature (start of shut-in), $T_{\text{initial}} = 140\text{ }^{\circ}\text{F}$
- Seabed ambient temperature, $T_{\text{amb}} = 40\text{ }^{\circ}\text{F}$
- Wax Appearance Temperature, $\text{WAT} = 100\text{ }^{\circ}\text{F}$ (Section 1.1 basis)
- Required no-flow protection time (Section 1.3 basis) = 6 hours
- Pipe: 8-in OD, 7.625-in ID; fluid density $\rho = 50\text{ lb/ft}^3$, specific heat $C_p = 0.5\text{ Btu/(lb}\cdot^{\circ}\text{F)}$
- Base-case insulation overall heat transfer coefficient, $U = 1.5\text{ Btu/(hr}\cdot\text{ft}^2\cdot^{\circ}\text{F)}$

Step 1 — Fluid heat capacity per unit pipe length:

$$A_{\text{cross}} = (\pi/4) \times (7.625/12)^2 = (\pi/4) \times (0.6354)^2 \approx 0.317\text{ ft}^2$$

$$\text{Mass per ft} = \rho \times A_{\text{cross}} = 50 \times 0.317 \approx 15.85\text{ lb/ft}$$

$$\text{Heat capacity per ft, } mC_p = 15.85 \times 0.5 \approx 7.92\text{ Btu/(ft}\cdot^{\circ}\text{F)}$$

Step 2 — Heat transfer area per unit length (based on OD):

$$A_{\text{area}} = \pi \times (8.625/12) \approx 2.259\text{ ft}^2/\text{ft}$$

Step 3 — Overall UA per unit length and thermal time constant:

$$UA = U \times A_{\text{area}} = 1.5 \times 2.259 \approx 3.39\text{ Btu/(hr}\cdot\text{ft}\cdot^{\circ}\text{F)}$$

$$\tau = mC_p / UA = 7.92 / 3.39 \approx 2.34\text{ hr}$$

Step 4 — Time to cool from T_{initial} to WAT (lumped-capacitance exponential decay):

$$(T(t) - T_{\text{amb}}) / (T_{\text{initial}} - T_{\text{amb}}) = \exp(-t/\tau)$$

$$(100 - 40) / (140 - 40) = 60/100 = 0.6$$

$$t = -\tau \times \ln(0.6) = 2.34 \times 0.5108 \approx 1.20\text{ hours}$$

Result: With the base-case insulation ($U = 1.5$), the fluid cools to WAT in only **≈ 1.2 hours** — far short of the **6-hour** required no-flow protection time. **FAIL.**

Step 5 — Back-calculate the required U to meet the 6-hour target:

$$\tau_{\text{required}} = t_{\text{target}} / (-\ln 0.6) = 6 / 0.5108 \approx 11.75\text{ hr}$$

$$UA_{\text{required}} = mC_p / \tau_{\text{required}} = 7.92 / 11.75 \approx 0.674\text{ Btu/(hr}\cdot\text{ft}\cdot^{\circ}\text{F)}$$

$$U_{\text{required}} = UA_{\text{required}} / A_{\text{area}} = 0.674 / 2.259 \approx 0.30\text{ Btu/(hr}\cdot\text{ft}^2\cdot^{\circ}\text{F)}$$

Result: Insulation must be upgraded to achieve **$U \approx 0.30\text{ Btu/(hr}\cdot\text{ft}^2\cdot^{\circ}\text{F)}$** — roughly **5× better** than the base case — to meet the 6-hour no-flow protection target. This is consistent with the thicker syntactic-foam or pipe-in-pipe insulation systems commonly used on real

subsea tiebacks, and confirms why insulation selection is a first-order cost/schedule driver rather than a minor detail.

✦ **Assumption check:** This lumped-capacitance method ignores pipe wall thermal mass and axial conduction/burial effects — it is a valid **first-pass screening** calc, but final insulation selection should be verified with a full transient thermal simulation (OLGA or equivalent) that also captures the temperature profile along the full pipeline length, not just a single representative point.

8.4 Calc Sheet 4 — Erosional Velocity Screening (API RP 14E)

Given:

- API RP 14E constant, $C = 100$ (continuous service, sand present, per Section 1.3 basis)
- Mixture density at flowing conditions, $\rho_m = 25 \text{ lb/ft}^3$
- Pipe: 6-in ID (0.5 ft)
- Volumetric flow at flowing conditions, $Q = 25,000 \text{ ft}^3/\text{hr} = 6.94 \text{ ft}^3/\text{s}$

Step 1 — Erosional velocity limit:

$$V_e = C / \sqrt{\rho_m} = 100 / \sqrt{25} = 100 / 5 = 20 \text{ ft/s}$$

Step 2 — Actual velocity in the 6-in line:

$$A = (\pi/4) \times (0.5)^2 = 0.1963 \text{ ft}^2$$
$$V_{\text{actual}} = Q / A = 6.94 / 0.1963 \approx 35.4 \text{ ft/s}$$

Step 3 — Compare:

$$V_{\text{actual}} (35.4 \text{ ft/s}) > V_e (20 \text{ ft/s}) \rightarrow \text{FAIL} - \text{erosion risk}$$

Step 4 — Required pipe area/diameter to meet the erosional limit:

$$A_{\text{required}} = Q / V_e = 6.94 / 20 = 0.347 \text{ ft}^2$$
$$D_{\text{required}} = \sqrt{4 \times A_{\text{required}} / \pi} = \sqrt{0.4415} \approx 0.664 \text{ ft} \approx 7.97 \text{ in}$$

Result: The 6-inch line exceeds the API RP 14E erosional velocity limit; upsizing to **~8-inch ID** brings velocity within the guideline.

✦ **Assumption check:** API RP 14E is a **screening** guideline, not a rate-of-metal-loss prediction — a line that "passes" the C-factor check can still erode over time if sand production is higher than assumed. For any non-trivial sand rate, follow up with a

quantitative erosion model (e.g., DNV RP O501) at the specific fittings/bends of concern, not just the straight-run check shown here.

8.5 Calc Sheet 5 — CO₂ Corrosion Rate Screening (de Waard-Milliams)

Given:

- Flowing temperature, $T = 60\text{ }^{\circ}\text{C} = 333\text{ K}$
- Total pressure = 50 bar, CO₂ content = 5 mol% → partial pressure of CO₂, $p(\text{CO}_2) = 0.05 \times 50 = 2.5\text{ bar}$
- Simplified de Waard-Milliams correlation (screening only — does not include pH, flow-regime, or glycol/corrosion-inhibitor correction factors used in the full model)

Step 1 — Apply the correlation:

$$\log(\text{CR}) = 5.8 - 1,710/T - 0.67 \times \log[p(\text{CO}_2)]$$

$$1,710 / 333 = 5.135$$

$$\log(2.5) = 0.398$$

$$0.67 \times 0.398 = 0.267$$

$$\log(\text{CR}) = 5.8 - 5.135 - 0.267 = 0.398$$

$$\text{CR} = 10^{0.398} \approx 2.50\text{ mm/year}$$

Result: Uninhibited CO₂ corrosion rate $\approx 2.5\text{ mm/year}$ — high, and unacceptable against a 3 mm total corrosion allowance over a 25-year design life without mitigation.

Step 2 — Apply design corrosion inhibitor efficiency (90%, Section 1.3 basis):

$$\text{CR}_{\text{inhibited}} = \text{CR} \times (1 - 0.90) = 2.5 \times 0.10 = 0.25\text{ mm/year}$$

Step 3 — Compare cumulative loss against corrosion allowance:

$$\text{Cumulative loss over 25 years} = 0.25 \times 25 = 6.25\text{ mm}$$

$$\text{Corrosion allowance} = 3\text{ mm}$$

$$6.25\text{ mm} > 3\text{ mm} \rightarrow \text{FAIL, even with a 90\% efficient inhibitor}$$

Result: Even with a 90%-efficient corrosion inhibitor, the projected cumulative loss (6.25 mm) **exceeds** the 3 mm corrosion allowance over the design life. Mitigation options: qualify a higher-efficiency inhibitor ($\geq 95\%$, giving 3.13 mm — still marginal), use corrosion-resistant alloy (CRA) cladding/lining for at least the highest-risk sections (e.g., near the choke where the largest pressure/temperature change occurs), or increase the base wall thickness/corrosion allowance.

✦ **Assumption check:** This screening correlation is uncorrected for pH, flow regime, and protective scale formation, all of which the full de Waard-Milliams/NORSOK M-506 models account for — treat this as a **first-pass red-flag screening tool**, not a final material-selection basis. A rich CO₂/temperature combination that screens this poorly should always trigger a full corrosion study before finalizing material selection.

8.6 Calc Sheet 6 — Scale Formation Screening (Saturation Ratio)

Given:

- Produced water Ca²⁺ concentration = 0.01 mol/L
- Produced water CO₃²⁻ concentration (from alkalinity/pH) = 2×10^{-5} mol/L
- Solubility product of CaCO₃ at reservoir/flowline temperature, $K_{sp} \approx 3.3 \times 10^{-9}$

Step 1 — Calculate the ion product (IP):

$$IP = [Ca^{2+}] \times [CO_3^{2-}] = 0.01 \times 2 \times 10^{-5} = 2 \times 10^{-7}$$

Step 2 — Calculate the saturation ratio (SR):

$$SR = IP / K_{sp} = (2 \times 10^{-7}) / (3.3 \times 10^{-9}) \approx 60.6$$

Result: SR ≈ **60.6** — since SR ≫ 1, the water is **severely supersaturated** with respect to CaCO₃, confirming a strong scaling tendency and the need for continuous scale inhibitor injection (or squeeze treatment) at the predicted precipitation location (commonly the choke or other high pressure-drop point where CO₂ flashing raises pH and drives CaCO₃ precipitation).

✦ **Assumption check:** This simplified ion-product screening ignores activity coefficients/ionic strength effects, which matter more at high total dissolved solids (TDS) — for produced/seawater-mixing systems with high TDS, use a full geochemical model (Stiff-Davis Index for high-salinity brines, or software such as ScaleChem/PHREEQC) rather than this simplified check for the final inhibitor-selection basis. This calc is a good, fast **red-flag screening tool**, not a design-basis-grade result.

9. Sample Datasheets

9.1 Chemical Injection (Methanol / LDHI) Datasheet

Parameter	Value	Unit
Tag No.	CIP-301 (Methanol Injection Package)	—
Service	Hydrate Inhibition — Continuous & Batch	—
Chemical	Methanol (MeOH), 99.9%	—
Design Injection Rate (early-life)	26 (per Calc Sheet 8.2)	bbl/day
Design Injection Rate (late-life, 40% WC)	To be recalculated per Calc Sheet 8.2 method	bbl/day
Storage Capacity	14 days at late-life rate	—
Injection Pressure	3,800 (above max wellhead pressure + margin)	psig
Injection Points	Subsea tree (downhole/wellhead), umbilical-fed	—
Pump Type	Triplex positive displacement, duty/standby	—
Turndown	10:1	—
Materials of Construction	316L SS (methanol-compatible wetted parts)	—
LDHI Backup System	Yes — anti-agglomerant, for moderate-subcooling normal operation	—
Applicable Standard	API RP 14E (piping), project chemical injection philosophy	—

9.2 Pipeline Thermal Insulation Datasheet

Parameter	Value	Unit
Line No.	FL-201 (Wellhead to Riser Base)	—
Nominal Size	8	in
Insulation Type	Multi-layer syntactic polyurethane foam, pipe-in-pipe	—
Design Overall U-value	0.30 (per Calc Sheet 8.3)	Btu/(hr·ft ² ·°F)
Required No-Flow Protection Time	6	hours
Arrival Temperature (steady-state, design flow)	140	°F
Ambient (Seabed) Temperature	40	°F
WAT	100	°F
Hydrate-Curve Temperature (at line pressure)	65	°F
Active Heating System	Electrical trace heating, standby (beyond 6-hr passive protection)	—
Insulation Design Life	25	years
Applicable Standard	Project thermal design philosophy, DNV-RP-F113 (deepwater pipeline insulation)	—

9.3 Corrosion & Erosion Monitoring Datasheet

Parameter	Value	Unit
System	Flowline FL-201, Wellhead to Host	—
CO ₂ Partial Pressure (design)	2.5 (per Calc Sheet 8.5)	bar
Uninhibited Corrosion Rate (screening)	2.5	mm/year
Design Inhibitor Efficiency (required)	≥ 95% (per Calc Sheet 8.5 result)	%
Corrosion Allowance	3	mm
Material — General Flowline	Carbon steel + corrosion inhibitor	—
Material — High-Risk Sections (choke, near-wellhead)	CRA-clad (e.g., 316L or Alloy 825 liner)	—
Erosional Velocity Limit (C=100 basis)	20 (per Calc Sheet 8.4)	ft/s
Governing Line Size (post-erosion check)	8 (upsized from 6-in)	in
Monitoring Method	ER probes + intelligent pigging (every 3 years)	—
Sour Service Screening	Required — H ₂ S = 50 ppm, per NACE MR0175/ISO 15156	—

10. Practical Design Checklist

- ☐ Flow assurance design basis issued and approved (Section 1) before thermal/chemical injection sizing begins
- ☐ Hydrate equilibrium curve generated from compositional thermodynamic model (not a generic chart) — see Calc Sheet 8.1
- ☐ Subcooling margin calculated at all cold-spot locations, including late-field-life water-cut case
- ☐ Methanol/MEG or LDHI dosage sized for both early-life and late-life water cut — see Calc Sheet 8.2
- ☐ WAT and pour point confirmed by lab testing on representative fluid, not assumed from analog fields
- ☐ Steady-state AND transient (no-flow cooldown) insulation performance checked — see Calc Sheet 8.3

- ☐ Asphaltene onset pressure (AOP) envelope checked against the full pressure-depletion production trajectory, not just initial conditions
 - ☐ Produced/injection water compatibility and scale saturation ratio screened — see Calc Sheet 8.6
 - ☐ CO₂/H₂S corrosion basis established; sour service material screening (NACE MR0175/ISO 15156) completed if H₂S present
 - ☐ Corrosion inhibitor efficiency requirement checked against corrosion allowance and design life — see Calc Sheet 8.5
 - ☐ Erosional velocity checked at all line sizes and high-risk fittings (bends, chokes) — see Calc Sheet 8.4
 - ☐ Slugging behavior modeled (OLGA/PIPESIM) and slug catcher sized against predicted volume/frequency
 - ☐ Restart/shut-in hydrate management plan developed (depressurization philosophy, inhibitor pre-treatment, or both)
 - ☐ Datasheets (chemical injection, insulation, corrosion/erosion monitoring) issued for vendor/EPC use
-

11. Common Field Issues & Lessons Learned

Issue	Root Cause	Practical Fix
Hydrate plug formed during unplanned shutdown, despite "adequate" insulation on paper	Insulation was sized against steady-state arrival temperature only, not the transient no-flow cooldown time	Always run the transient cooldown check (Calc Sheet 8.3) against the actual required no-flow protection time, not just steady-state
Methanol storage/pump undersized within a few years of startup	Dosage sized only for early-life water cut	Size chemical injection systems for the late-life (highest water cut) case, per Calc Sheet 8.2 note
Unexpected wax buildup at a riser base rather than at the coldest seabed point	Deposition assumed to track ambient temperature alone; local flow regime/shear and cooling rate also matter	Use full thermal + deposition-rate simulation along the entire profile, not just the single coldest point
Asphaltene deposition found in tubing/choke despite "wax control" measures in place	Asphaltene and wax were treated as the same problem with the same controls	Recognize AOP is pressure/chemistry driven, not temperature driven — chase the pressure-depletion trajectory against the AOP envelope, not the temperature profile
Scale-related choke plugging shortly after seawater injection started	Produced/injection water compatibility not tested before commissioning	Run compatibility and saturation-ratio screening (Calc Sheet 8.6) before committing to any water injection scheme
Corrosion-related leak at a bend well before design life	Screening corrosion rate used but inhibitor efficiency assumed rather than lab-qualified, and erosion at the bend was stripping the inhibitor film	Lab-qualify actual inhibitor efficiency; check erosion and corrosion together at high-risk fittings, not independently

12. Case Study — Subsea Tieback Hydrate Blockage on Emergency Shutdown

A composite, illustrative case study based on the type of finding commonly encountered on subsea gas-condensate tiebacks following an unplanned shutdown. Names, tag numbers, and figures are representative, not project-specific.

12.1 Background

A subsea gas-condensate tieback (the illustrative project used throughout this guide) had been in operation for approximately 4 years. The original flow assurance design basis sized pipeline insulation and methanol injection using the early-life water cut (5%) and a 6-hour no-flow protection time, consistent with the project's original planned-shutdown philosophy. Methanol injection had never been required in normal operation, since steady-state arrival temperature stayed comfortably above WAT and the hydrate curve.

12.2 Problem Identified

An unplanned host platform trip caused an **8-hour** unplanned shutdown of flowline FL-201 — exceeding the original 6-hour design basis. By this point in field life, water cut had risen to approximately 35%, close to the late-life design case (40%), though the methanol system had never been re-verified against this updated water cut.

On restart, operations identified a **partial hydrate blockage** at the riser base, evidenced by an abnormal pressure differential across that section and a failed pigging attempt. Production was shut in for approximately 9 days while the blockage was managed (a combination of controlled depressurization from both ends and extended methanol circulation) before flow was safely re-established.

12.3 Investigation & Recalculation

The flow assurance team reran the cooldown and subcooling analysis using this guide's Calc Sheet 8.1 and 8.3 methodology:

- **Cooldown check (Calc Sheet 8.3 method):** Using the as-built insulation U-value against an 8-hour (not 6-hour) shut-in duration confirmed that fluid temperature at the riser base had fallen below the hydrate equilibrium temperature at the shut-in (static) pressure roughly **1.5 hours before** the platform trip was cleared and restart was attempted — the fluid had been in the hydrate-risk region for over 6 hours before restart procedures began.
- **Subcooling check (Calc Sheet 8.1 method):** At the actual shut-in pressure and the calculated riser-base temperature, subcooling was approximately **22 °F** — well beyond the small margin the original methanol philosophy assumed would be available "if needed," since the design basis had never actually required active methanol dosing during a credible shutdown scenario.
- **Root finding:** The design basis's 6-hour no-flow protection assumption had never been stress-tested against a realistic unplanned-shutdown duration distribution — 6 hours was the *planned* maintenance shutdown target, not a validated bound on all credible unplanned trips.

12.4 Root Cause

Two compounding root causes were identified:

1. **Design basis scope gap** — the 6-hour no-flow protection target was carried over from the planned-maintenance philosophy without a separate check against unplanned/emergency shutdown duration statistics (which, per the host platform's own trip history, included multiple prior events exceeding 6 hours).
2. **No active hydrate management procedure for shutdowns beyond 6 hours** — the operations team had a restart procedure for routine cases but no pre-established, pressure-tested depressurization/methanol-circulation procedure for an extended unplanned shutdown, costing valuable time during the actual event.

12.5 Resolution

- The flow assurance design basis was revised to size passive insulation (or add active heating) against the **credible unplanned shutdown duration** (revised to a 24-hour basis, based on the host platform's trip history) rather than only the 6-hour planned-maintenance target.
- A pre-approved **extended shutdown hydrate management procedure** was developed and drilled with operations, covering controlled depressurization targets and methanol circulation rates, so a future event would not require real-time engineering analysis under time pressure.
- The methanol system's dosage basis (Calc Sheet 8.2 methodology) was re-run against the current 35% water cut and confirmed adequate for pre-emptive dosing ahead of a planned/forecasted extended shutdown, but was flagged as **not** sized for continuous protection through an unlimited-duration unplanned event — reinforcing that depressurization, not indefinite methanol dosing, is the primary strategy for long unplanned shutdowns.

12.6 Outcome

- The 9-day production deferral had a significant, quantifiable revenue impact and prompted a company-wide review of flow assurance design-basis assumptions across other subsea assets with similar "planned-maintenance-only" no-flow protection targets.
- The finding was documented as a corporate lessons-learned item: no-flow protection time targets must be validated against **actual credible shutdown duration statistics** for the specific host facility, not adopted as a generic industry rule of thumb.

12.7 Case-Specific Lessons Learned

Lesson	Applied Fix
A "planned maintenance" no-flow protection time target is not automatically valid for unplanned shutdown scenarios	Size insulation/heating against the host facility's actual credible unplanned-shutdown duration statistics, not just the planned-maintenance case
Methanol dosage basis must be re-verified as water cut rises through field life	Schedule periodic (e.g., annual) re-verification of Calc Sheet 8.2-style dosage checks against current water cut, not just at initial design
Time pressure during a real event leads to worse outcomes than a pre-drilled procedure	Develop and drill an extended-shutdown hydrate management procedure (depressurization targets, circulation rates) before it is needed
Depressurization, not indefinite chemical dosing, is the practical strategy for long unplanned shutdowns	State this explicitly in the design basis and operating philosophy so operations does not default to "just inject more methanol" as an open-ended strategy

13. Reference Standards

- **API RP 14E** — Recommended Practice for Design and Installation of Offshore Production Platform Piping Systems (erosional velocity guideline)
- **NORSOK M-506** — CO₂ Corrosion Rate Calculation Model
- **NACE MR0175 / ISO 15156** — Petroleum and natural gas industries — Materials for use in H₂S-containing environments in oil and gas production (sour service)
- **DNV-RP-F113** — Pipeline Subsea Repair (referenced alongside general deepwater pipeline insulation/thermal design practice)
- **DNV RP O501** — Erosive Wear in Piping Systems
- Hammerschmidt, E.G. (1934) — original correlation for hydrate suppression by inhibitor concentration
- de Waard, C. & Milliams, D.E. (1975) — Prediction of Carbonic Acid Corrosion in Natural Gas Pipelines

This guide is a practical study reference combining standard flow assurance design theory with worked sample calculations and lessons learned from real project experience. All numeric examples are illustrative — always validate against project-specific PVT/water-chemistry data, dynamic simulation results (OLGA/PIPESIM), and current regulatory/code requirements.